

SMART INDIA HACKATHON 2026

CogniCare NER

AI-Based Cognitive Gaming & Memory
Assistance Platform for Elderly Dementia

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01

Executive Summary

CogniCare NER is a highly specialized, AI-powered cognitive gaming and memory assistance platform engineered explicitly for elderly dementia patients in the North Eastern Region (NER). Built for the Ministry of Development of North Eastern Region (MDoNER), this platform directly tackles the massive rural healthcare deficit by deploying culturally localized, offline-first digital therapy.

At its core, CogniCare replaces generic, English-centric puzzle apps with highly tailored cognitive treatments. It utilizes **Gemini 3.5 AI** for empathetic regional language processing, **Web Audio API** for procedural folk music therapy (binaural beats), and a proprietary **Vocal Biomarker Analyzer** that converts natural speech cadence into clinical MMSE-grade reports for doctors. Crucially, the entire system is architected as an offline-first **Progressive Web App (PWA)**, guaranteeing continuous therapy even in zero-connectivity rural zones.

Live Clinical Prototype

The complete frontend architecture is successfully deployed. It features 13 interactive cognitive games, 7 NER languages, real-time Vocal Biomarker processing via Gemini, and 100% Edge/Offline capability via Service Workers.

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Problem Statement & Deep Analysis

Dementia affects over 5.3 million elderly in India, with the NER facing extreme compounding challenges. Generic cognitive apps (like Lumosity) fail spectacularly in this demographic for three critical reasons:

Critical Barrier	Impact on Current Healthcare System
The Language & Literacy Barrier	Existing apps rely on English UI and complex text instructions. Rural NER elderly (Assamese, Mizo, Khasi speakers) are immediately alienated by text-heavy interfaces.
The Rural Connectivity Deficit	Mainstream ML therapy apps demand persistent 4G connections for cloud-AI processing. In rural NER, internet is intermittent, rendering cloud-dependent apps useless.
Cultural & Semantic Disconnect	Dementia therapy requires triggering long-term autobiographical memory. Matching generic triangles or recognizing American landmarks does nothing for a patient in Majuli; they need familiar, regional stimuli.
The Clinical Tracking Void	Caregivers lack the medical training to objectively track decline, leading to reactive emergency care rather than proactive management. Doctors receive no continuous telemetry data.

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Proposed Solution (8 Core Pillars)

CogniCare NER systematically dismantles every barrier mentioned above through 8 interconnected, highly engineered technical pillars:

- **1. AI Cognitive Gaming:** 13 distinct clinical games targeting 8 specific cognitive domains (Memory, Executive Function, Visuospatial, etc.).
- **2. Adaptive Difficulty Engine:** A real-time ML algorithm that monitors patient struggle (time-to-click, error rate) and auto-scales game complexity to prevent frustration.
- **3. Voice-First NER Interface:** Powered by Gemini 3.5 Assistant, allowing the elderly to navigate entirely by speaking in Assamese, Manipuri, Khasi, Mizo, Bengali, Hindi, or English.
- **4. Caregiver Clinical Dashboard:** Aggregates raw gameplay data and telemetry, converting it into an MMSE-grade Cognitive Wellness Index (CWI) score with automated PDF reports for doctors.
- **5. 100% Offline Edge PWA:** Utilizes IndexedDB and Service Workers to run the entire game engine, telemetry tracking, and procedural audio directly on the device CPU without the internet.
- **6. NER Cultural Therapy:** Games are deeply rooted in regional geography. Patients complete Bihu proverbs, identify Kaziranga animals, and listen to synthesized NER folk instruments to trigger deep memory.
- **7. Smart Reminders & SOS:** Automates caregiver duties with audio medicine/hydration reminders spoken in the native dialect, alongside a one-tap Emergency SOS for wandering risks.
- **8. Geriatric-First UI Design:** Engineered strictly for failing eyesight and motor control. Features 56px+ touch targets, single-tap navigation, ultra-high contrast, and zero nested menus.

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System Architecture & Edge Flow

The platform is built on a 4-tier edge-native architecture designed to minimize server latency and maximize local processing.

Architecture Tier	Technical Components
1. Patient Interface	Web Speech API for STT voice capture, touch events, and a highly accessible Geriatric UI.
2. Core AI Engine	Gemini 3.5 Voice Assistant, Adaptive Difficulty heuristics, and the Vocal Biomarker Analyzer.
3. Data & Sync Layer	IndexedDB for persistent offline caching, Service Workers for asset delivery, Web Audio API.
4. Clinical Output	Caregiver Analytics Dashboard, Sundowning Alerts, MMSE Progress Reports, Medicine Reminders.

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Core AI Engine & Vocal Biomarkers

5.1 Vocal Biomarker Detection

The most advanced clinical feature of CogniCare is its ability to conduct passive cognitive assessments via voice. When the patient speaks to the AI Assistant, the system does not just transcribe the text; it analyzes the way the patient speaks.

Vocal Biomarker	Clinical Significance
Type-Token Ratio (TTR)	Measures vocabulary richness. A dropping TTR indicates semantic memory loss (forgetting words).
Filler Word Frequency	Counts excessive use of "um," "uh," and regional equivalents, indicating high cognitive load.
Speech Cadence & Pauses	Measures milliseconds of silence between words. Increased hesitation strongly correlates with early-stage dementia.
Sentiment Fluctuations	Detects signs of agitation or depression, which are common precursors to Sundowning syndrome.

5.2 Gemini 3.5 Integration

We leverage Google Gemini 3.5 Flash for rapid, empathetic conversational AI. The prompt engineering is specifically tuned to behave as a patient, reassuring companion (Smriti Saathi) that never corrects the patient aggressively, thereby avoiding catastrophic reactions common in dementia care.

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100% Offline Edge Infrastructure

To solve the rural connectivity crisis, we completely bypassed traditional cloud-heavy app architecture. CogniCare is a Progressive Web App (PWA) that

acts as a standalone local application.

Edge Technology	Implementation Details
Service Workers	Intercepts all network requests. Caches the React bundle, CSS, and game logic locally upon first load. The app opens instantly even in airplane mode.
IndexedDB Storage	Instead of requiring MongoDB for every action, all game telemetry, CWI scores, and voice logs are saved locally in the browser's IndexedDB. Background Sync API pushes data to the cloud only when WiFi returns.
Procedural Web Audio	Instead of downloading hundreds of megabytes of MP3 files for music therapy, we use the browser's native Web Audio API (Oscillators, Gain Nodes) to mathematically synthesize NER instruments on the fly. Zero bandwidth required.

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Platform Modules (13 Clinical Games)

The therapy engine comprises 13 mini-games mapped directly to the 8 standard clinical cognitive domains.

1. Memory Match (Cultural)

Short-Term Memory: Patients match pairs of Bihu Dhols, Rhinos, and local textiles. Time-to-match is heavily tracked.

2. Family Faces (Emotional)

Autobiographical Memory: Caregivers upload family photos. The system asks "Where is Rahul?" to reinforce immediate family bonds.

3. Melody Memory (Auditory)

Working Memory: Simon-says style pattern matching using synthesized NER instruments (Pepa, Pung) instead of standard beeps.

4. Daily Routine Ordering

Executive Functioning: Drag-and-drop chronological sorting of daily tasks (e.g., Wake Up -> Tea -> Medicine).

5. Cultural Proverb Completion

Language Preservation: Completes famous regional proverbs. Helps maintain semantic vocabulary.

6. Spatial Shape Rotation

Visuospatial Skills: Mentally rotating traditional NER weaving patterns to fit into a grid.

7. Sundowning Audio Therapy

Clinical Module: Generates 40Hz Gamma binaural beats mixed with procedural rain and folk drone to combat evening agitation. (Not a game, but a therapy tool).

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Cultural Integration (NER Therapy)

Clinical research proves that dementia patients respond best to stimuli rooted in their deep past. CogniCare replaces sterile, clinical shapes with vibrant, culturally resonant NER themes:

- **Visuals:** Kaziranga Safari themes, Hornbill festival aesthetics, and regional attire.
- **Audio:** Authentic procedural synthesis of the Bihu Dhol, Bamboo Flute, and Mizo Gong.
- **Language:** Medicine reminders are delivered in the precise local dialect (e.g., Assamese, Khasi), which feels like a family member speaking rather than a robotic alarm.

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Technology Stack

Engineered for maximum performance on low-end Android devices.

System Layer	Technologies Used
Frontend / UI	React 18, Vite (Rapid Compilation), Tailwind/Custom CSS, Lucide Icons.
Edge / Offline	PWA Service Workers (Workbox), IndexedDB (LocalForage), Web Audio API.
Backend / Sync	Node.js, Express.js REST API, MongoDB Atlas (for eventual cloud sync).
AI Capabilities	Google Gemini 3.5 Flash, Web Speech API (Native STT/TTS).

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Feasibility, Viability & Security

Factor	Explanation
Zero Hardware Cost	Runs natively in Chrome/Edge on existing ■5,000 Android phones. No specialized medical tablets required.
Frictionless Distribution	As a PWA, users simply visit the URL and click "Add to Home Screen". Bypasses the complex Google Play Store update process.
Patient Data Privacy	Since AI processing and data storage happen locally via IndexedDB, sensitive medical telemetry never leaves the device unless explicitly authorized by the caregiver for cloud backup.

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Clinical Impact & Benefits

- Social Benefit:** The Voice-First UI in 7 regional languages removes the digital barrier, restoring dignity to the elderly.
- Economical Benefit:** Free decentralized PWA saves rural families thousands of rupees in traditional therapy travel costs.
- Environmental Benefit:** Edge-AI and procedural audio consume near-zero battery, reducing cloud server carbon footprints and e-waste.
- Strategic Benefit:** Early detection of cognitive decline via Vocal Biomarkers allows MDoNER to proactively triage medical resources across districts.

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Competitive Advantage

Feature	CogniCare NER	Lumosity / Elevate
7 Regional NER Languages	Yes	No (English only)
Vocal Biomarker Tracking	Yes	No
Procedural Folk Audio Therapy	Yes	No
100% Offline PWA (No internet)	Yes	No (Requires 4G)
MMSE-Grade Doctor Reports	Yes	No

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Implementation Roadmap

Phase	Objective
Phase 1 (Months 1-2)	Clinical Validation: Pilot testing the 13 games with 50 local NER patients to fine-tune the Adaptive Difficulty algorithm.
Phase 2 (Months 3-4)	Language Expansion: Finalizing the Voice-UI for Khasi, Mizo, and Manipuri dialects with native voice-actors for TTS.
Phase 3 (Months 5-6)	MDoNER Rollout: Deploying the PWA through local primary healthcare centers and ASHA workers in rural districts.

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Team Members

#	Team Member Name
1	Darshan Jain (Team Leader)
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-- End of Technical Report --